

ReactJS Hands-on - Conditional Rendering (bloggerapp)

Book Details, Blog Details and Course Details using different conditional rendering techniques

1. Objective

Create a React app **bloggerapp** with three components - **Book Details**, **Blog Details** and **Course Details** - implemented using as many different ways of conditional rendering as possible: **if / else**, the **ternary operator**, the **&& operator**, and list rendering with **map()** and **keys**.

2. Prerequisites

- Node.js
- NPM
- Visual Studio Code

3. Create the React App

● ● ● Command Prompt

```
C:\Users\student> npx create-react-app bloggerapp
Creating a new React app in C:\Users\student\bloggerapp.
Installing react, react-dom, and react-scripts...

Success! Created bloggerapp at C:\Users\student\bloggerapp
Happy hacking!

C:\Users\student> cd bloggerapp
C:\Users\student\bloggerapp> code .
```

4. Mock Data

src/books.js

```
export const books = [
  { id: 101, bname: 'Master React', price: 670 },
  { id: 102, bname: 'Deep Dive into Angular 11', price: 800 },
  { id: 103, bname: 'Mongo Essentials', price: 450 }
];
```

src/courses.js

```
export const courses = [
  { id: 201, cname: 'React', duration: '40 hrs' },
  { id: 202, cname: 'Angular', duration: '35 hrs' },
  { id: 203, cname: 'MongoDB', duration: '20 hrs' }
];
```

5. Book Details - List Rendering with map() and Keys

The book list is built once with `Array.map()` and stored in a variable; each item gets a unique key so React can track it efficiently.

```
const bookdet = (
  <div>
    {books.map((book) => (
      <div key={book.id}>
        <h3>{book.bname}</h3>
        <h4>{book.price}</h4>
      </div>
    ))}
  </div>
);
```

6. Blog Details - Conditional Rendering with if / else

A plain `if / else` inside a function chooses which JSX to return, based on a flag.

```
function getBlogContent() {
  if (isLoggedIn) {
    return (
      <div>
        <h3>React Learning</h3>
        <p>React is a JavaScript library for building user interfaces.
          You can install React from npm.</p>
      </div>
    );
  } else {
    return <p>Please login to view blogs.</p>;
  }
}
```

```
}  
}  
const content = getBlogContent();
```

7. Course Details - Ternary Operator and && Operator

The ternary operator picks between the course list and a fallback message; the && operator conditionally renders an extra note only when the condition is true.

```
const coursedet = courses.length > 0 ? (  
  <ul>  
    {courses.map((course) => (  
      <li key={course.id}>{course.cname} - {course.duration}</li>  
    ))}  
  </ul>  
) : (  
  <p>No courses available</p>  
)  
);  
  
{/* && operator: renders only when the condition is true */}  
{courses.length > 5 && <p>Showing top courses only</p>}
```

8. App.js - Putting It Together

src/App.js

```
import React from 'react';  
import { books } from './books';  
import { courses } from './courses';  
import './App.css';  
  
function App() {  
  const isLoggedIn = true;  
  
  function getBlogContent() {  
    if (isLoggedIn) {  
      return (  
        <div>  
          <h3>React Learning</h3>  
          <p>React is a JavaScript library for building user interfaces.  
            You can install React from npm.</p>  
        </div>  
      );  
    } else {  
      return <p>Please login to view blogs.</p>;  
    }  
  }  
  
  const content = getBlogContent();  
  
  const coursedet = courses.length > 0 ? (  
    <ul>  
      {courses.map((course) => (  
        <li key={course.id}>{course.cname} - {course.duration}</li>  
      ))}  
    </ul>  
  ) : (  
    <p>No courses available</p>  
  );  
  
  const bookdet = (  
    <div>  
      {books.map((book) => (  
        <div key={book.id}>  
          <h3>{book.bname}</h3>  
          <h4>{book.price}</h4>  
        </div>  
      ))}  
    </div>  
  );  
  
  return (  
    <div className="App">  
      <div className="st2">  
        <h1>Book Details</h1>  
        {bookdet}  
      </div>  
      <div className="v1">  
        <h1>Blog Details</h1>  
        {content}  
      </div>  
      <div className="mystyle1">  
        <h1>Course Details</h1>  
        {coursedet}  
        {courses.length > 5 && <p>Showing top courses only</p>}  
      </div>  
    </div>  
  );  
}
```

```
    </div>
  );
}

export default App;
```

9. Run the Application

● ● ● Command Prompt

```
C:\Users\student\bloggerapp> npm start
Compiled successfully!

You can now view bloggerapp in the browser.
Local: http://localhost:3000
```

10. Output

● ● ● localhost:3000

Book Details

Master React

670

Deep Dive into Angular 11

800

Mongo Essentials

450

Blog Details

React Learning

React is a JavaScript library for building user interfaces. You can install React from npm.

Course Details

- React - 40 hrs
- Angular - 35 hrs
- MongoDB - 20 hrs

isLoggedIn is set to true, so the if/else in Blog Details renders the article; courses.length is 3 (> 0), so the ternary in Course Details renders the list instead of "No courses available"; the && note is skipped since there are only 3 courses, not more than 5.

11. Observation

The same outcome – deciding what to render – can be reached with an if/else function, a ternary expression, or the && operator; which one to use is mostly a matter of how much logic is involved and how readable the JSX stays. map() combined with a stable key on each item is the standard way to render a dynamic list and lets React update only the rows that actually changed.